Trustworthy Machine Learning, 2026
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Task: Watermark Forgery Attack

The goal of watermark forgery is to implement a watermark forging scheme achieving a high false detection rate (bit accuracy) while maintaining high image quality. Watermark forgery attacks, in which a watermark is stolen from genuine content and applied to malicious content, present serious potential for harm. Your goal as a group is to successfully launch such attacks under (almost) black box conditions.

As the attackers, you have access to:

- **Target Images:** 200 clean images (in the `clean_targets` folder), which serve as the target for your forgery attacks.

- **Source Images:** 8 unidentified watermarking methods, each providing a dataset of 25 watermarked images (`WM_1` through `WM_8`) all with the same message from the `watermarked_sources` folder.

What you don't have access to:

- Information about the watermarking schemes that were applied to the images.

What's your task:

Your mission is to subtly alter the target images in such a way that they bear the watermark embedded in the respective set of source images. We expect you to submit the images with the forged watermark.

Target Mapping: You must apply the extracted watermarks to the clean images according to the following exact batches:

  - `WM_1` → Clean images `1.png` to `25.png`
  - `WM_2` → Clean images `26.png` to `50.png`
  - `WM_3` → Clean images `51.png` to `75.png`
  - `WM_4` → Clean images `76.png` to `100.png`
  - `WM_5` → Clean images `101.png` to `125.png`
  - `WM_6` → Clean images `126.png` to `150.png`
  - `WM_7` → Clean images `151.png` to `175.png`
  - `WM_8` → Clean images `176.png` to `200.png`

This logic is also implemented in task_template.py.

Your Starting Point

To start you off, we provide a task_template.py that shows you how to load the dataset and save your submission zip in the required manner. We also provide a submission.py that handles submitting your model to the evaluation server.

All data files can be found at HuggingFace.

Evaluation:

Your submission is scored based on a combination of the **Strength** of the forgery and the **Visual Quality** of the resulting image.

1. Detection Strength (Sdet): The strength of the forgery is measured by the bitwise accuracy of the recovered watermarking message. It is scaled such that random guessing (50% bit-accuracy) yields a score of 0, and perfect forgery yields a score of 1:

$$S_{det} = max(0.0, (BitAccuracy - 0.5) * 2)$$

2. Visual Quality (Sqlt): The visual quality is measured using **LPIPS** (Learned Perceptual Image Patch Similarity). Because LPIPS measures perceptual distance, it aligns closely with human perception. The score decays exponentially as the image degrades:

$$S_{qlt} = e^{(-8 * LPIPS)}$$

3. Final Score: Your final score per image is the product of your detection strength and your visual quality. You must succeed at both to score points.

$$S_{final} = S_{det} * S_{qlt}$$

How to submit your results?

- We have provided a unique API-key to every team. You are supposed to only use this token to submit your results to our server.
- **Your submission must be a .zip file containing the 200 clean images with forged watermarks. No other artifacts or subfolders allowed.** File names must remain **unaltered**. Do not include subfolders. You are supposed to submit your solution using the code provided in submission.py.
- Remember to replace YOUR_API_KEY_HERE with your actual API token (keep the double quotes), replace PATH/TO/YOUR/ZIP.zip with the path to your solution. Also set SUBMIT to True if not already.
- To avoid the possibility of brute-forcing, only one submission can be made every 60 minutes. If the submission was unsuccessful due to an error, this cooldown period will only be 2 minutes long.

Scoring
All submissions are automatically validated and evaluated on the hidden test set.

- **Public leaderboard:** based on 30% of samples

- **Private leaderboard:** final ranking based on the remaining 70%

This means that **optimizing only the leaderboard score does not automatically give you a good final result**. Your results must generalize for all models. The leaderboard score reflects the score achieved by the submission. Higher is better.

Leaderboard
Results are published at: http://34.63.153.158/leaderboard_page
The leaderboard shows the best result per team only. If your current submission score is lower than the score saved in the leaderboard, the score will not be updated.

Points to note regarding submission:
You can only submit a file (zip) with scores **every 60 minutes**. If your submission is malformed or wrong, you should get a comprehensive error message. However, even if you fail, you get a **2 minute** cooldown.

Read the following instructions carefully:
Additionally, you are required to submit a **ZIP file** to CMS that contains your code and a report that explains your approach to the task. This file should contain:

1. Report
- Matriculation number and **CMS team Id** in the report
- An Introduction:
    - A short paragraph explaining **your understanding of the task** written by your own words.
- A Main body:
    - Explaining **your approach to the task** including the details about the hyperparameters and structural choices you have made, as well as the reasons you made that choice.
    - Explaining **key results** that you have obtained on the public leaderboard.
- A Conclusion
    - In your conclusion try to be goal oriented. In other words, try to answer this question: now that you successfully forged watermarks, what practical implications does this vulnerability have in real-world systems and why should we be concerned?

2. Code Repository
- Containing all your code files, well documented and clean
- Every experiment mentioned in the report must have corresponding code in the repository
- **Avoid including large model artifacts and virtual environments in your repository.**
- Add a README.md that ONLY explains how to **recreate** your best leaderboard result
- You are only supposed to make one submission per group

The report should NOT be more than **2 pages** long. The submitted zip file that does not contain the code and/or the report will NOT be considered for evaluation. For the report you

SHOULD use the template for ICLR on overleaf. Note that there is no need for an abstract and you should skip it.

The submission to the leaderboard will end automatically on **07.07.2026 23:59** and after that you will not be able to submit any results to the leaderboard. The same date also applies to CMS. As a result, make sure to submit your results and report on time. Thereafter, you will be able to check your evaluation results performed on the held-out set on the leaderboard. No late submissions will be accepted via email (no exceptions).

References:

- P. Yang, H. Ci, Y. Song and M. Z. Shou, "Can Simple Averaging Defeat Modern Watermarks?" in Advances in Neural Information Processing Systems (NeurIPS), vol. 37, December 2024
- Z. Dong, C. Shuai, Z. Ba, P. Cheng, Z. Qin, Q. Wang and K. Ren, "WMCopier: Forging Invisible Image Watermarks on Arbitrary Images" in Advances in Neural Information Processing Systems (NeurIPS), vol. 38, December 2025
- T. Souček, S. Rebuffi, P. Fernandez, N. Jovanović, H. Elsahar, V. Lacatusu, T. Tranand A. Mourachko, "Transferable Black-Box One-Shot Forging of Watermarks via Image Preference Models" in Advances in Neural Information Processing Systems (NeurIPS), vol. 38, December 2025

Note - When reading papers about watermark forging, keep an eye on the dataset and compute requierments of each method. You don't have to immediately discard methods which use more than you have of either, they might still contain helpful information.

- S. Craver, N. Memon, B. . -L. Yeo and M. M. Yeung, "Resolving rightful ownerships with invisible watermarking techniques: limitations, attacks, and implications" in *IEEE Journal on Selected Areas in Communications*, vol. 16, no. 4, pp. 573-586, May 1998
- M. Kutter and S. Voloshynovskiy and A. Herrigel, "The Watermark Copy Attack" in *Security and Watermarking of Multimedia Contents II* (SPIE Proceedings), vol. 397, pp. 371-380, January 2000